



ZAHIN UKASYAH

BACHELOR IN SOFTWARE ENGINEERING
@UNIVERSITI TEKNOLOGI MALAYSIA



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Maluri, Kuala Lumpur



Software Engineering student specializing in AI and NLP. Proven ability in model optimization, including LLM quantization (GGUF, vLLM) and Agentic RAG pipeline development. Strong foundation in scalable full-stack development (React, Next.js) to deliver high-performance, integrated intelligent applications.

SKILL

FULLSTACK DEVELOPMENT

- **Core Technologies:** HTML, CSS, JavaScript (JS), TypeScript (TS), Tailwind CSS, API integration.
- **Frameworks & Libraries:** React, Next.js, Node.js, ShadCN UI, Laravel.
- **Mobile Development:** Expo, Tunnel.
- **Database:** Supabase, MongoDB, PHP.
- **Backend:** RESTful API, FastAPI.
- **UI Shell:** v0, Replit, Lovable, Bolt
- **Automation:** n8n, OpenAI Builder.
- **Deployment:** Vercel, Railway, Server.
- **IDE/ CLI:** Cursor, Warp, Claude Code.
- **Operating System:** Linux, Windows.

DEVOPS & TOOLS

- **Version Control:** Git, GitHub, GitLab.
- **CI/CD:** Docker, Kubernetes, Redis.

DATA SCIENCE & MACHINE LEARNING

- **Core Languages & Libraries:** Python, pandas, NumPy, scikit-learn, Keras.
- **Deep Learning:** CNN, Fuzzy Logic, Regression, Genetic Algorithm.
- **AI:** Agentic RAG, Semantic Search, Quantization, Vector Embedding, Inference Server, Reranker.
- **Data Acquisition:** Beautiful Soup.
- **IDE/ CLI:** Google Colab, JupyterNB.

LANGUAGE PROFICIENCY

- **Bahasa Melayu:** Native Speaker.
- **English:** Professional Proficiency.

EXPERIENCE

AI/NLP ENGINEER INTERN (R&D) | MIMOS BERHAD, MOSTI.

PROJECT: MALAYSIA MENTAL HEALTH MOBILE APPLICATION

- Accelerated real-time conversational flow by reducing LLM inference latency by 3.5× through migrating the serving pipeline from **llama.cpp** to the **vLLM** framework.
- Ensured scalable application accessibility by reducing the core model memory footprint over 60% by implementing a **4-bit quantization**, enabling efficient user-side deployment.
- Improved the factual integrity of therapeutic guidance by integrating a multi-document **Retrieval-Augmented Generation (RAG)** pipeline that retrieves and grounds responses in Clinical Manuals and Research Papers.
- Designed and implemented the end-to-end **Agentic RAG** pipeline, resulting in a 40% reduction in hallucinated responses by enforcing a confidence-scoring mechanism.
- Enhanced long-term user support by developing a **memory feature** that retains and indexes key personal details.

RESEARCH ASSISTANT | UNIVERSITI TEKNOLOGI MALAYSIA

PROJECT: CNN+LSTM FOR AI-POWERED DEEPFAKE DETECTION

- Accelerated model evaluation and research cycles by reducing testing time by 40% through developing and automating a standardized deep learning evaluation script using **Python** and **TensorFlow/PyTorch**.
- Validated the efficacy of hybrid architectures by testing **CNN+LSTM** models against known deepfake datasets, providing quantitative performance metrics (F1-score, AUC) to inform the final model selection.
- Contributed foundational academic knowledge by conducting comprehensive systematic literature reviews on state-of-the-art deepfake detection and temporal sequence modeling techniques to support the paper submission.

REFERENCE

DR. HALINAWATI BINTI HIROL

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System Electronic Engineering,
Faculty Malaysia-Japan International
Institute of Technology,
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EXTRACURRICULAR

SOFTWARE ENGINEERING SOCIETY | DEC 2024 - NOV 2025

- **President:** Effectively led the organization, skillfully oversaw strategic to achieve organizational goals.

STUDENT REPRESENTATIVE COUNCIL | DEC 2023 - DEC 2024

- **Exco Student Welfare:** Actively advocated for student welfare and successfully organized support initiatives.



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PROJECTS

IRIS BIOMETRIC DETECTION (FYP) | UTM PROJECT

- Accomplished a highly accurate iris recognition system with 99.8% accuracy on the CASIA-v4 dataset using a **CNN** and ArcFace loss.
- Developing a deep learning model for iris recognition using **Python** and **TensorFlow**.
- Integrating a **ResNet50** and the ArcFace loss function to improve recognition performance.

FAKE NEWS DETECTION MODEL | UTM PROJECT

- Accomplished the automated classification of news articles with an accuracy of 94.6% on a publicly available Kaggle dataset.
- Developing and training a machine learning model in **Python** using **scikit-learn** and **NLTK**.
- Implementing a complete data pipeline, including efficient data loading with **pandas** and feature extraction using **TF-IDF**.

STUDENT MERIT SYSTEM | SK. INTEGRASI AL-ALUSI

- Accomplished a significant reduction in administrative overhead for the Student Merit & Demerit System as measured by optimizing the staff and student registration process.
- Developing a custom web application using **Ruby** to replace a manual, paper-based workflow.
- Integrating **Supabase** for the student records.

MOODCAST: WEATHER TRACKER | UTM PROJECT

- Built a data-driven weather application that provides real-time weather information by fetching data from **WeatherMap RESTful API**.
- Leveraged **TypeScript** to enforce type safety by creating custom interfaces for the complex.
- Utilized **React's** **useEffect** hook to manage API calls effectively, triggering data fetches based on user input and component lifecycle events.

INVOLVEMENT

MAYBANK FOUNDATION

- Academic Scholar (Current):** Awarded for academic excellence and leadership potential.
- Ambassador (2024):** Represented Maybank to promote its initiatives by participated in Maybank events by organizer.

SOFTWARE ENGINEERING SOCIETY UTM

- President (2025):** Led the society and oversee organized events to benefit community.
- Director (2025):** Organized and directed society Annual Grand Meeting 2025 for the interim.
- Secretary (2024):** Managed records and facilitated communication in the organization.
- Liaison Officer (2024):** Coordinate Flow and Protocol for handling Guests in events.

PEMBIMBING RAKAN SISWA UTM

- Acting President (2024):** Led the organization and ensured smooth operations.
- Vice President (2024):** Assisted in leadership, planning, and organizational management.
- Director (2023):** Organized student development workshop for Event: Student Character Building.
- Exco Protocol (2023):** Managed protocols and coordinated with external stakeholders.

MIMOS BERHAD, MOSTI

- AI/NLP Engineer (Current):** Engineered the full-stack AI platform, efficiently optimizing core **NLP** models (LLMs, RAG) and integrated client-side infrastructure (React/Next.js) for the Project.

SOFTWARE ENGINEERING DEPARTMENT UTMKL

- FYP Research (Current):** Studied biometric image detection using **CNN**, on iris recognition with the CASIA-IrisV4 dataset, including data scraping and model testing in **Python** (TensorFlow, Keras)
- Research Assistant (2025):** Conducted a research review on **CNN+LSTM** models for AI-Powered Deepfake Detection and testing using **Python**.

STUDENT REPRESENTATIVE COUNCIL UTM

- Exco Student Welfare (2024):** Advocated for student welfare and organized support initiatives.
- Director (2024):** Initiate the Scholarship Hunter project to aid student on financial issues.
- Deputy Director (2024):** Organized the planning for Event: Minggu Mesra Mahasiswa 2024.
- Deputy Director (2024):** Organized the planning for Event: Eid Al-Fitr Celebration UTMKL 2024.
- Liaison Officer (2024):** Facilitated communication with China's delegates on UTMKL business travel.